

Thermal Camera

Wiring and setup guide for a Pimoroni MLX90640, a Raspberry Pi 2 Model B and an Adafruit PiTFT 3.5" touchscreen. The camera app starts by itself at every boot.

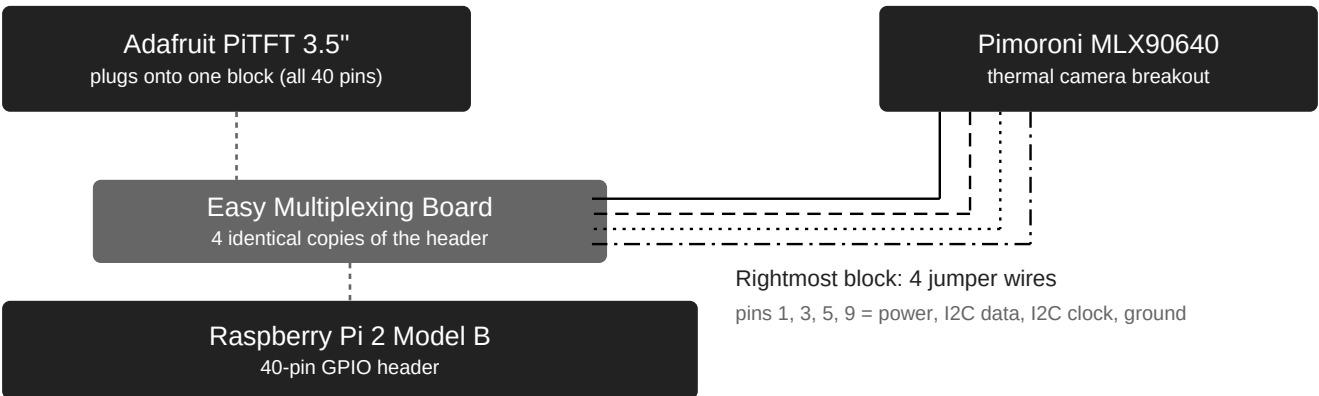
Updated 26 September 2026, after the first build on real hardware. Matches the code at github.com/marco-ponds/thermal-camera.

1. Parts

Part	Job	How it connects
Raspberry Pi 2 Model B	Runs the camera app	Power bank via micro-USB
Easy Multiplexing Board (52Pi)	Four identical copies of the 40-pin header	Stacks on the Pi's header
Adafruit PiTFT 3.5" (480x320, resistive touch)	Screen and touch input	Plugs onto one block of the multiplexing board (SPI)
Pimoroni MLX90640 breakout	32x24 pixel thermal camera	4 jumper wires to the rightmost block (I2C)
4 female-female jumper wires	Sensor wiring	Sensor pins to the multiplexing board
SanDisk 32GB microSD	Operating system and app	Pi's microSD slot
USB Wi-Fi dongle	Network for setup, SSH and copying snapshots	Any USB port
Power bank + short micro-USB cable	Power: 5 V from a port rated 2 A or more (section 9)	Pi's micro-USB power socket

2. How it fits together

The screen and the camera use different buses, so they never compete for pins. The PiTFT uses SPI plus two control pins; the camera uses I2C on pins 3 and 5, which the PiTFT leaves free. The multiplexing board repeats the whole header four times, so every block has the same pins in the same places.

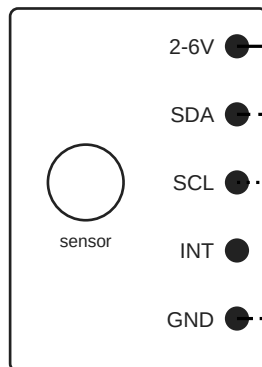


Always wire with the power off. Unplug the power bank before connecting or moving jumper wires or the screen. A wire slipping onto a 5V pin while powered is the easiest way to damage the sensor or the Pi.

3. Wiring the camera

Use the **rightmost block** of the multiplexing board. Its printed labels read, from the top, **3.3V, SDA, SCL, GCLK, GND**, and they belong to the column of pins right next to them (the odd pins 1, 3, 5, 7, 9). Wire the 1st, 2nd, 3rd and 5th pins of that column. Each wire is drawn with its own line style so the drawing still reads in greyscale; the colours are only suggestions.

Pimoroni MLX90640 (pins as printed)



red wire: 2-6V to 3.3V (pin 1)

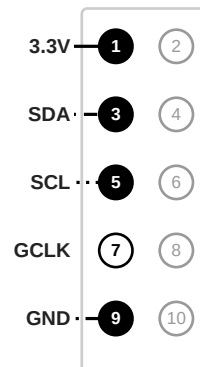
blue wire: SDA to SDA (pin 3)

yellow wire: SCL to SCL (pin 5)

INT: leave unconnected

black wire: GND to GND (pin 9)

Multiplexing board, rightmost block



top of the board = pin 1

Sensor pin	Board label	Pi pin	Wire (line)	Notes
2-6V	3.3V	1	Red (solid)	Always 3.3V, never 5V: with 5V the sensor's I2C lines would sit at 5V, which can damage the Pi.
SDA	SDA	3	Blue (dashed)	GPIO2, I2C data
SCL	SCL	5	Yellow (dotted)	GPIO3, I2C clock
INT				Not needed. Leave it unconnected.
GND	GND	9	Black (dash-dot)	Any GND pin works (6 and 9 are nearest). It must be a GND pin: on the first build this wire was on a GPIO pin and the sensor didn't show up at all.

Using a different block

Every block carries the same pins, but the labels sit beside different columns:

- **Rightmost block:** the labelled column (next to 3.3V / SDA / SCL / GCLK / GND).
- **Leftmost block:** the **outer, unlabelled** column. The labels printed beside it (5V, 5V, GND, TXD0...) belong to its inner column, the even pins, which the sensor must not use.
- **Middle blocks:** the column with 3.3V, IO2, IO3, IO4, GND next to it.

If you're unsure, check with a multimeter before connecting the sensor: with the Pi on, the top pin of the right column reads about 3.3 V against a GND pin. 5 V means you're on the wrong column.

Checklist before powering on

- The PiTFT covers a whole 2x20 block, the right way round, not shifted by a row.
- The red wire is on 3.3V (pin 1), not on a 5V pin.
- The black wire is on a pin labelled GND.
- The pin names printed on your breakout match the table. If they differ, go by name (2-6V, SDA, SCL, GND), not by position.

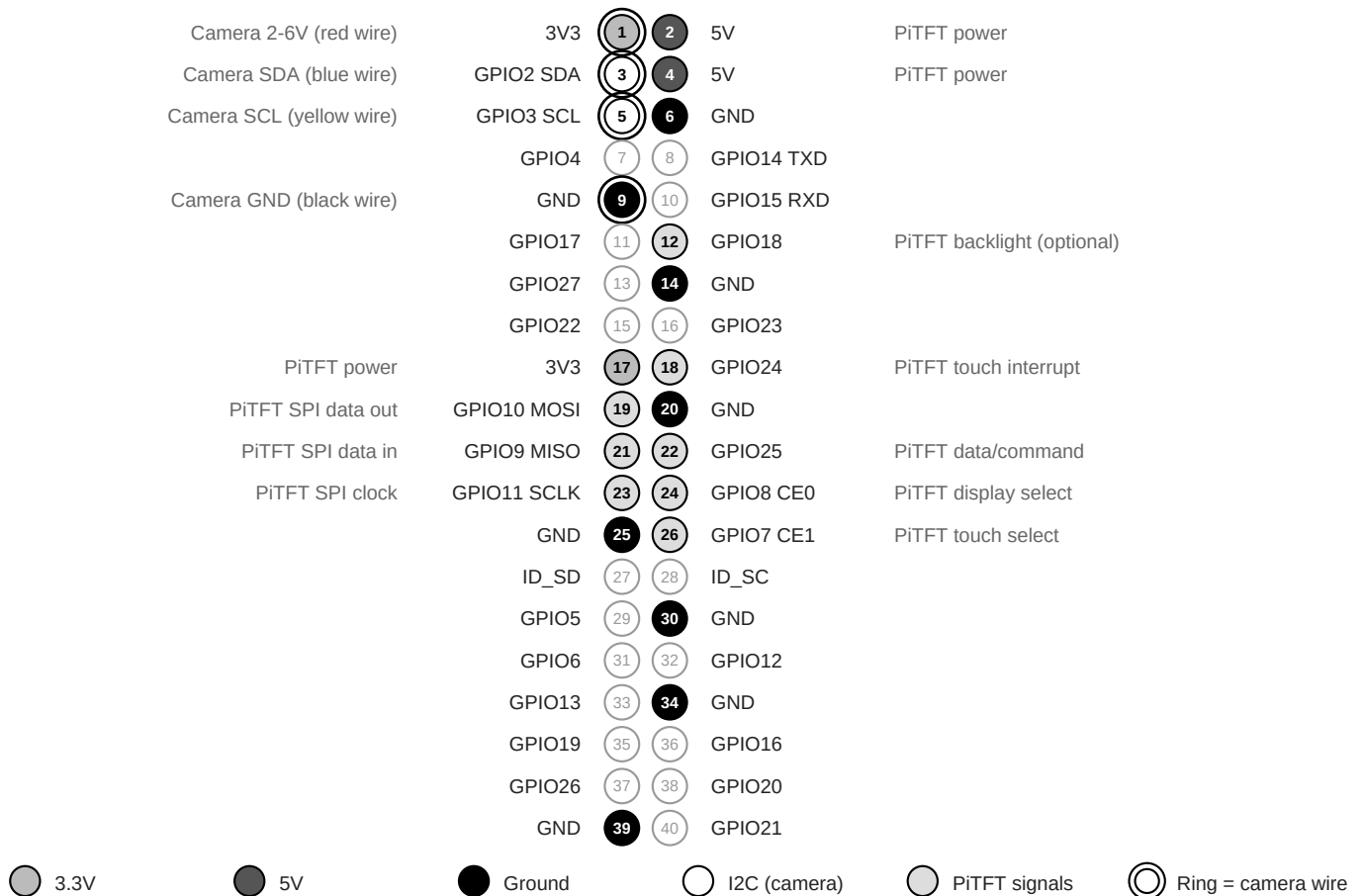
Mounting the sensor

The sensor sees a 4:3 landscape picture, like the screen. Mount it so that, holding the camera upright, the picture is upright. If you find yourself tilting the camera to see things the right way up, turn the sensor board on its mount by the same amount (on the first build that was 90° clockwise). The software can mirror the picture but not rotate it.

4. Full header map

Every pin of the 40-pin header, shaded by what uses it. The PiTFT plugs straight on, so you never wire its pins by hand; this map shows there's no overlap with the camera.

Odd pins in the left column, even pins in the right, pin 1 at the top



Pin(s)	Signal	Used by	Purpose
3, 5	GPIO2 SDA, GPIO3 SCL	Camera	I2C bus (sensor address 0x33)
19, 21, 23	GPIO10 MOSI, GPIO9 MISO, GPIO11 SCLK	PiTFT	SPI bus for display and touch
24 / 26	GPIO8 CE0 / GPIO7 CE1	PiTFT	Display chip select / touch controller chip select
22	GPIO25	PiTFT	Display data/command line
18	GPIO24	PiTFT	Touch interrupt
12	GPIO18	PiTFT (optional)	Backlight dimming, only if you enable it
1, 17 / 2, 4 / GND	3.3V / 5V / ground	Both	Power pins are shared; that's normal

PiTFT pins are from Adafruit's published pinout for the 3.5" resistive PiTFT.

5. Software setup

You do this once. Steps 1 and 2 happen at the bench; the rest runs over SSH from your laptop. Replace <user> with the username you choose in step 1.

1 Flash the microSD card

In Raspberry Pi Imager choose device **Raspberry Pi 2**, operating system **Raspberry Pi OS Lite (32-bit)** (under "Raspberry Pi OS (other)"), and your SD card. In OS customisation set:

- Hostname: **thermalcam**
- A username and password
- Your Wi-Fi name and password, wireless country GB
- Enable SSH with password authentication

2 Assemble and power on

With everything unplugged: multiplexing board on the Pi, PiTFT on one block, camera wired to the rightmost block (section 3), Wi-Fi dongle and SD card in. Then connect the power bank.

- The PiTFT stays **plain white** until its driver is installed in step 5. That's expected.
- **First boot takes 2 to 5 minutes** and the Pi reboots itself once. Later boots take 30 to 60 seconds.
- Red LED steady = power OK. Green LED flickering = reading the SD card; mostly dark = booted. A regular repeating flash pattern is an error code (usually a bad SD card).

3 Connect over SSH

From your laptop. The white screen tells you nothing, so check over the network:

```
ping thermalcam.local          # Ctrl+C once it answers
ssh <user>@thermalcam.local
```

If ping never answers after 5 minutes, it's usually the Wi-Fi details in Imager or the dongle. Your router's list of connected devices shows whether the Pi joined.

4 Get the code

On the Pi:

```
sudo apt-get update
sudo apt-get install -y git
git clone https://github.com/marco-ponds/thermal-camera.git
cd thermal-camera
```

5 Run the setup script

```
./setup.sh
```

Run it as your normal user (not with sudo); it asks for your password once. It takes 15 to 30 minutes on a Pi 2 and ends with "Done. Reboot now". It:

- installs pygame, numpy, evdev, i2c-tools and the DejaVu fonts
- creates the Python environment with Blinka and the MLX90640 library
- installs the **PiTFT screen driver** with Adafruit's installer (`--display=35r --rotation=90 --install-type=drivers`)
- loads the **touchscreen driver** (`stmpe_ts`) now and at every boot
- turns on I2C at 400 kHz
- installs the **thermalcam** service, which starts the app at boot and keeps the text console off the screen

Why not HDMI mirror mode? Older instructions (including an earlier version of this guide) used `--install-type=mirror`. Adafruit's installer refuses that on current Raspberry Pi OS Lite, so the driver is installed on its own and the app draws straight onto the screen.

6 Reboot and calibrate the touchscreen

```
sudo reboot
```

After about a minute the PiTFT shows **three crosses, one at a time**. Tap the centre of each firmly with a fingernail or stylus. Then the Live view appears. From now on the app starts by itself at every boot.

7 Check it works

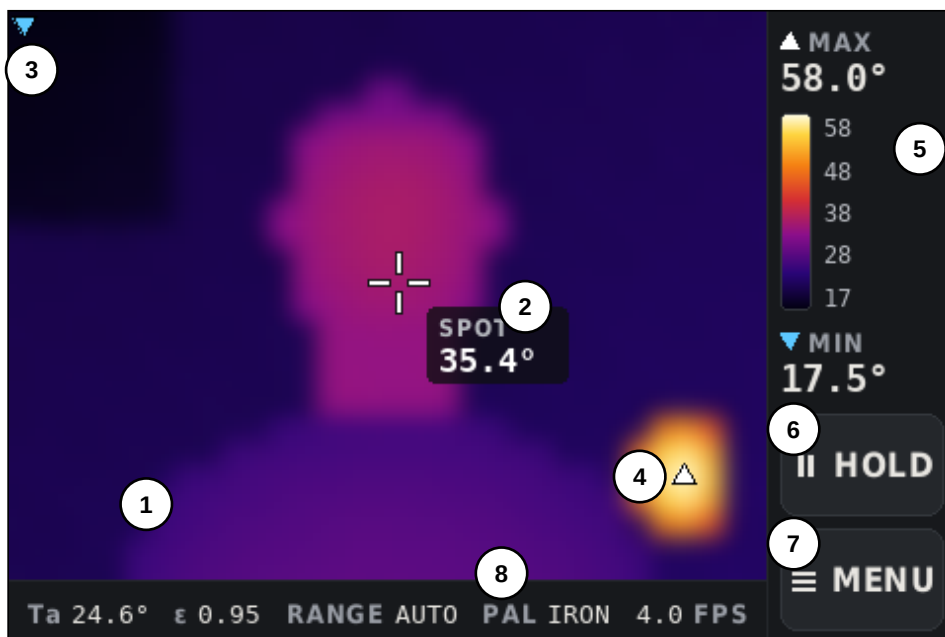
SSH back in and run:

```
i2cdetect -y 1          # the grid shows 33 = sensor found
systemctl status thermalcam # "active (running)"
journalctl -u thermalcam -b | tail # which screen and touchscreen it found
vcgencmd get_throttled    # throttled=0x0 = power is fine
```

If anything is off, see section 10.

6. Using the camera

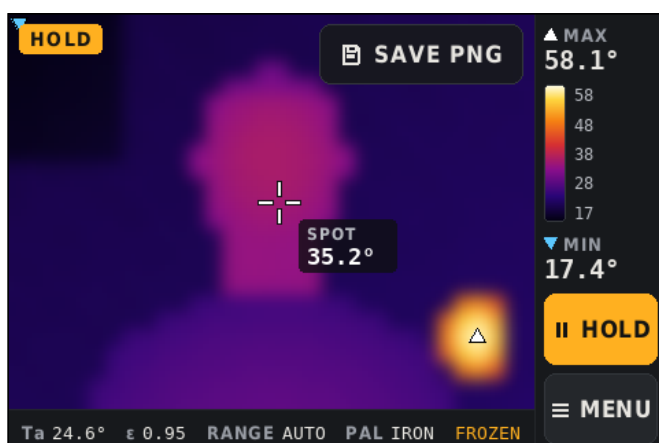
Live view



What it is

- | | |
|-------|---|
| 1 | The thermal image. Tap anywhere on it to move the spot meter to that pixel (it's remembered). |
| 2 | Spot meter: crosshair and SPOT reading. |
| 3 / 4 | Coldest point (down-pointing triangle) and hottest point (up-pointing triangle). |
| 5 | MAX and MIN of the whole frame, with the colour scale and 5 ticks in between. |
| 6 | HOLD: freeze the picture. Tap again to go live. |
| 7 | MENU: open Settings. |
| 8 | Strip: Ta (sensor chip temperature), e (emissivity), RANGE (AUTO or LOCK), PAL (palette) and frames per second. |

Hold and save



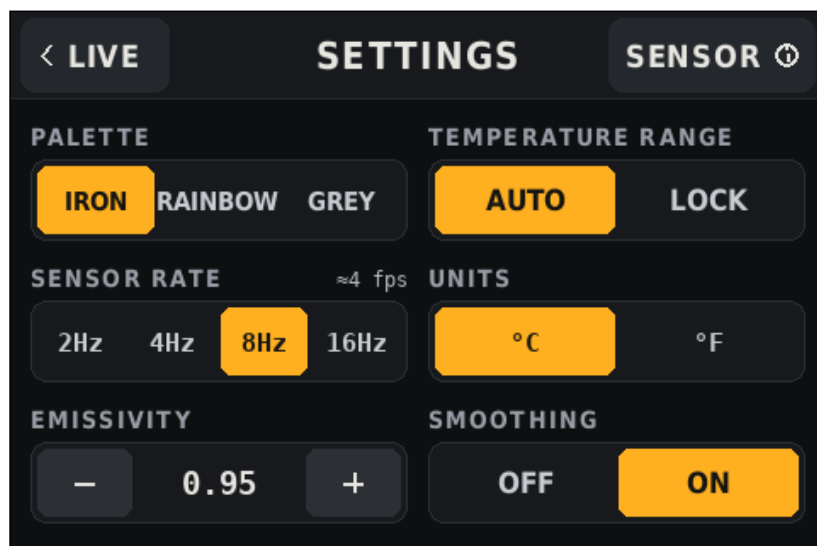
While held, the image is frozen, a **HOLD** badge shows top-left, the strip reads **FROZEN**, and a **SAVE PNG** button appears.

SAVE writes two files to
~/thermalcam/snapshots/:

- YYYYMMDD-HHMMSS.png: the screen as shown
- ...-raw.csv: 24 rows x 32 columns of °C

The Pi 2 has no clock battery, so until it has synced the time over Wi-Fi the files are numbered snap-0001, snap-0002...

Settings



Every change applies immediately and is kept after a reboot. **SENSOR** (top right) opens Sensor info; **LIVE** (top left) goes back.

Setting	Options	What it does
Palette	Iron / Rainbow / Grey	The colour map. Grey reads best on e-ink screenshots.
Temperature range	Auto / Lock	Lock fixes the colour scale at the current min and max, so the same colour means the same temperature in every room.
Sensor rate	2 / 4 / 8 / 16 Hz	The sensor's half-frame rate; full images arrive at half this (shown as approx. N fps). A Pi 2 manages about 1 to 4 fps whatever you pick.
Units	°C / °F	Every temperature on every screen.
Emissivity	0.10 to 1.00	0.95 suits paint, plaster, wood, fabric and skin.
Smoothing	Off / On	Raw 32x24 blocks, or a smooth blurred picture.

Sensor info

< BACK		SENSOR INFO		LIVE 
HARDWARE		LIVE		
Model	MLX90640BAB	Die temp (Ta)	24.6 °C	
Field of view	55° × 35°	Supply (Vdd)	3.29 V	
Array	32 × 24	ADC resolution	18-bit	
Object range	−40...300 °C	Readout pattern	Chess	
I ² C address	0x33	Subpage rate	8 Hz	
Bus speed	400 kHz	Frame rate	4.0 fps	
Device ID	FAKE · 0000 · 0001	Frames / dropped	9 / 0	
Bad pixels	0 / 768	Pi CPU temp	—	

Refreshed about once a second. **BACK** returns to Settings, **LIVE** to the camera. Anything that can't be read shows a dash.

Field	Meaning
Model / Field of view	MLX90640BAB = 55° x 35° (Pimoroni's standard), BAA = 110° x 75° (wide). The sensor can't report its lens, so this comes from the config file.
Bus speed	The I2C clock actually in use (400 kHz).
Device ID	The sensor's unique serial number.
Bad pixels	Faulty pixels listed in the sensor's memory (up to 4 is within spec). They're filled in from their neighbours.
Die temp (Ta)	The sensor chip's own temperature. It sits above room temperature and isn't a scene reading.
Supply (Vdd)	The sensor's supply voltage. Close to 3.3 V is right.
ADC / Readout / Subpage	Read live from the sensor: ADC resolution, readout pattern and half-frame rate.
Frames / dropped	Frames read since the app started, and failed reads. A few dropped frames are normal; lots point to a loose wire.

7. Finding cold spots

- **Pick a cold day.** You need about a 10 °C difference between inside and outside for draughts and missing insulation to show clearly.
- **Lock the range in a normal room first**, then walk around. With auto range every room looks like it has a cold spot, because the colours stretch to fit whatever is in view.
- **Check the usual suspects:** window and door frames, skirting boards, where walls meet the ceiling, pipes and cables through walls, loft hatches, sockets on outside walls.
- **Don't trust glass or shiny surfaces.** Windows, mirrors, tiles and metal reflect heat from elsewhere (often you). Aim at the frame, not the pane.
- **Get close.** With 32x24 pixels each pixel covers more wall the further away you stand. Scan from 1 to 2 metres, and move the spot meter onto anything suspicious.

8. Everyday commands

On the Pi over SSH (ssh <user>@thermalmcam.local), unless noted.

To do this	Run
Shut down safely before unplugging	<code>sudo poweroff</code>
See the app's log	<code>journalctl -u thermalmcam -f</code>
Stop / start / restart the app	<code>sudo systemctl stop start restart thermalmcam</code>
Run it by hand (stop the service first)	<code>cd ~/thermal-camera && .venv/bin/python main.py</code>
Test the screen without the sensor	add <code>--fake-sensor</code> to the command above
Redo the touch calibration	add <code>--calibrate</code> to the command above
Update to the latest code	<code>cd ~/thermal-camera && git pull && SKIP_PITFT=1 ./setup.sh && sudo reboot</code>
Copy snapshots to your laptop (on the laptop)	<code>scp '<user>@thermalmcam.local:thermalmcam/snapshots/*' .</code>

Shut down before unplugging the power bank when you can. The settings file is written safely, but cutting power while the Pi writes is the usual cause of a corrupted SD card. Wait for the green light to stop flashing, then unplug.

The config file

Most settings are on the Settings screen. A few live only in `~/config/thermalmcam/config.json`. After editing it, run `sudo systemctl restart thermalmcam`.

Key	Default	Change it when
<code>flip_h / flip_v</code>	<code>true / false</code>	The picture is mirrored left-right / upside down
<code>sensor_model</code>	<code>"MLX90640BAB"</code>	You have the 110° wide-angle sensor ("MLX90640BAA")
<code>touch_min_pressure</code>	<code>0</code>	Light brushes trigger taps (try 20 to 40)
<code>touch_calibration</code>	<code>set on first boot</code>	Delete it, or run with <code>--calibrate</code> , to calibrate again

9. Power

Any USB power bank gives 5 V, which is what the Pi 2 takes through its micro-USB socket. What matters is the current: use a port rated **2 A or more**.

Part	Typical draw at 5 V
Pi 2 under load	400 to 600 mA
PiTFT (mostly the backlight)	about 100 mA
USB Wi-Fi dongle	100 to 250 mA
MLX90640	about 20 mA
Total	about 0.7 to 1 A, with short spikes higher

- **10,000 mAh** gives roughly 6 to 8 hours; 5,000 mAh about 3.
- USB-C power banks are fine with a USB-C to micro-USB cable.
- **The cable matters as much as the bank.** Thin or long cables drop voltage. Use a short (50 cm or less), thick one.
- Check with `vcgencmd get_throttled`: 0x0 is fine; anything else means the voltage dipped.

10. Troubleshooting

Symptom	Likely cause and fix
Screen stays plain white	The screen driver isn't installed yet. Run <code>./setup.sh</code> and reboot (step 5). If it's still white: check the PiTFT covers a whole block the right way round, and that <code>cat /sys/class/graphics/fb*/virtual_size</code> shows 480,320.
"SENSOR NOT FOUND - RETRYING" and i2cdetect shows an empty grid	Wiring. Check the GND wire is on a GND pin first (the cause on the first build), then that the wires are on the right column (section 3). About 3.3 V between the sensor's 2-6V and GND pins means it has power. The app reconnects by itself once the sensor answers.
No calibration crosses; taps do nothing	The touchscreen driver isn't loaded. Current <code>setup.sh</code> loads it; on an older install run: <code>sudo modprobe stmpe_ts</code> , then <code>echo stmpe_ts sudo tee /etc/modules-load.d/stmpe-ts.conf</code> , then <code>sudo systemctl restart thermalcam</code> .
Boot messages or terminal text cover the Live view	An older install. <code>cd ~/thermal-camera && git pull && SKIP_PITFT=1 ./setup.sh && sudo reboot</code> .
Taps land in the wrong place	Stop the service and run <code>main.py --calibrate</code> (section 8).
You have to tilt the camera to see things upright	Rotate the sensor on its mount (section 3).
Picture mirrored or upside down	Set <code>flip_h</code> or <code>flip_v</code> in the config file. For the whole screen upside down: <code>PITFT_ROTATION=270 ./setup.sh</code> , then reboot.
Many dropped frames on Sensor info	A loose or long jumper. Reseat the wires, keep them short, or choose a lower sensor rate.
Random reboots, or get_throttled isn't 0x0	Not enough power: a 2 A port and a short, thick cable (section 9).
Everything looks about the same temperature	Not enough difference indoors. Try a colder day, and use LOCK so small differences stay visible.